

A Deep Learning Framework for Translating Tamil Sign Language Gestures into Speech

PROJECT PHASE I REPORT

Submitted by

NIKILESH S **2116221801035**

POONGAVIN B **2116221801036**

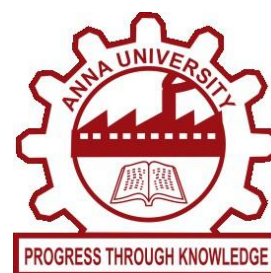
SUKISH M **2116221801053**

in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



RAJALAKSHMI ENGINEERING COLLEGE

(AUTONOMOUS), CHENNAI – 602 105

NOV 2025

BONAFIDE CERTIFICATE

Certified that this Report titled “**A DEEP LEARNING FRAMEWORK FOR TRANSLATING TAMIL SIGN LANGUAGE GESTURES INTO SPEECH**” is the bonafide work of “**NIKILESH S (2116221801035), POONGAVIN B (2116221801036) and SUKISH M (2116221801053)**” who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE**Dr. J.M. Gnanasekar M.E., Ph.D.,****Professor and Head,**Department of Artificial Intelligence
and Data Science

Rajalakshmi Engineering College

Thandalam – 602 105

SIGNATURE**Dr. A. BEULAH M. Tech.,
Ph. D.,****Asso. Professor,**Department of Artificial
Intelligence and Data ScienceRajalakshmi Engineering
College

Thandalam – 602 105

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- To demonstrate effective teamwork, project management, and communication skills through collaborative project execution.
- To instill awareness of ethical, societal, and environmental considerations in the design and deployment of intelligent systems.

COURSE OUTCOME

CO 1: Analyze and define a real-world problem by identifying key challenges, project requirements and constraints.

CO 2: Conduct a thorough literature review to evaluate existing solutions, identify research gaps and formulate research questions.

CO 3: Develop a detailed project plan by defining objectives, setting timelines, and identifying key deliverables to guide the implementation process.

CO 4: Design and implement a prototype or initial model based on the proposed solution framework using appropriate AI tools and technologies.

CO 5: Demonstrate teamwork, communication, and project management skills by preparing and presenting a well-structured project proposal and initial implementation results.

ABSTRACT

Communication barriers persist for the Tamil-speaking deaf and mute community, hindering their seamless interaction with the wider public, which necessitates a deep learning solution to bridge the gap between Tamil Sign Language (TSL) and spoken or written Tamil. The primary goal is to establish a viable and scalable deep learning framework for translating TSL gestures into corresponding text and speech, thereby enhancing communication accessibility. The framework employs MediaPipe Hand Landmark Detection, which identifies TSL gestures and extracts 21 precise hand key points from pre-collected gesture images. These extracted spatial features, capturing the nuances of the TSL alphabets, are then processed by a Convolutional Neural Network (CNN) for effective pattern learning and classification. The study aims to lay the groundwork for developing a robust and scalable TSL gesture-to-text system, representing an essential step toward future real-time applications that can significantly improve communication for the Tamil-speaking deaf and mute community.

Keywords – Tamil Sign Language (TSL), Sign Language Recognition (SLR), MediaPipe, Convolutional Neural Networks (CNN), Deep Learning, Hand Gesture Recognition, Real-Time Sign Recognition, Accessibility, Feature Extraction, Landmark Detection.

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S. MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. J.M. GNANASEKAR, M.E., Ph.D.**, Professor and Head of the Department of Artificial Intelligence and Data Science for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide, **Dr. A. BEULAH, M. Tech., Ph. D.**, Assistant Professor, Department of Artificial Intelligence and Data Science. Rajalakshmi Engineering College for her valuable guidance throughout the course of the project. We are very glad to thank our Project Coordinator, **Dr. S. SURESH KUMAR, M.E., Ph.D.**, Professor, Department of Artificial Intelligence and Data Science for his useful tips during our review to build our project.

NIKILESH S

(2116221801035)

POONGAVIN B

(2116221801036)

SUKISH M

(2116221801053)

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
ASL	American Sign Language
BSL	British Sign Language
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CSV	Comma-Separated Values
FPS	Frames Per Second
GPU	GPU – Graphics Processing Unit
ML	ML – Machine Learning
RGB	Red Green Blue
SDLC	Software Development Life Cycle
SLR	Sign Language Recognition
NLP	Natural Language Processing
TTS	Text-to-Speech
UI	User Interface
USA	United States of America
UK	United Kingdom

CHAPTER 1

INTRODUCTION

1.1 THE GLOBAL CHALLENGE OF HEARING IMPAIRMENT

Communication is the foundation of human interaction, education, employment, and social belonging. For millions of people around the world who are deaf or hard-of-hearing, this fundamental aspect of daily life becomes a major challenge. Hearing impairment not only affects an individual's ability to receive auditory information but also influences social participation, emotional well-being, and access to essential services such as healthcare, transportation, and education. Many individuals experience isolation, reduced employment opportunities, and significant communication barriers when interacting with the hearing community.

Historically, deaf communities across the world developed their own natural sign languages long before formal linguistic recognition. Early forms of sign communication were used in ancient civilizations, but modern sign languages began to develop systematically in the 18th and 19th centuries. For example, French Sign Language (LSF) played a major role in shaping American Sign Language (ASL), while British Sign Language (BSL) evolved separately despite sharing the same spoken English language. This shows that spoken language does not determine the structure of sign language—each sign language grows independently based on culture, geography, and community practices.

Today, sign languages are not universal. Different regions and countries have their own complete and linguistically rich sign languages, each with unique grammar, vocabulary, and handshape structures. Examples include ASL (USA), BSL (UK), ISL (Indian Sign Language), CSL (Chinese Sign Language), and thousands of regional variants. Within India itself, sign languages vary significantly across states, with Tamil Sign Language (TSL) being widely used in Tamil Nadu. These regional differences make communication even more challenging when communities interact outside their linguistic boundaries.

According to the World Health Organization (WHO), more than 430 million people worldwide require rehabilitation for disabling hearing loss, accounting for over 5% of the global population. As the number continues to rise due to aging populations, noise exposure, and lack of early intervention, the need for technological solutions becomes more urgent. There is a clear global demand for intelligent systems that can translate regional sign languages into spoken or written language, bridging the communication gap between the hearing and hearing-impaired communities. This necessity drives the development of modern computer-vision-based sign language recognition systems, especially for less-resourced languages like Tamil Sign Language.

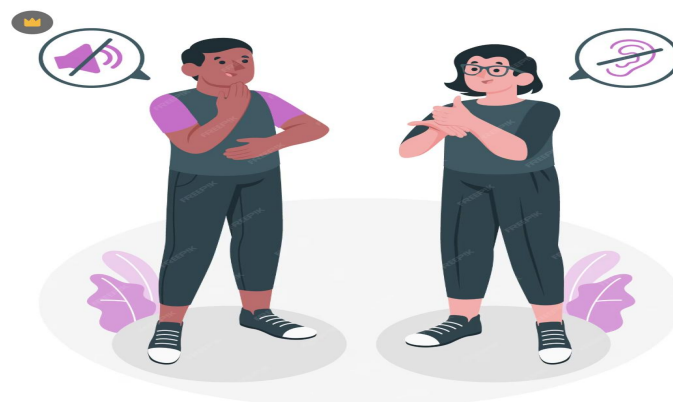


Figure 1: Deaf and Mute Community Representation [1]

1.2 THE SIGNIFICANCE OF TAMIL SIGN LANGUAGE (TSL)

Tamil Sign Language (TSL) is the primary mode of communication for thousands of deaf and mute individuals in Tamil Nadu and the global Tamil-speaking community. Although India has Indian Sign Language (ISL), TSL developed independently due to cultural and linguistic factors specific to Tamil society. Sign languages are not universal, and just like spoken Tamil differs greatly from other Indian languages, TSL also has its own grammar, gestures, and structure. This makes TSL an essential tool for social inclusion, education, and daily communication for the deaf community.

A major reason for TSL's uniqueness is its mapping to the Tamil script, which consists of 247 characters. Representing such a large and complex writing system

requires detailed handshapes, finger orientations, and combinations, unlike languages with fewer alphabets. Even small changes in gesture position or movement can alter the meaning completely. This complexity makes TSL significantly different from both ISL and Western sign languages like ASL or BSL. As a result, technologies built for other sign languages cannot directly work for Tamil sign recognition without region-specific adaptation.

Because of its linguistic richness and cultural importance, accurate recognition of TSL is essential for real-world applications. Many deaf individuals consider TSL their first and most natural language, and without proper technological support, they face barriers in education, employment, and public communication. Developing AI models that understand the unique structure of TSL not only preserves its cultural identity but also bridges major accessibility gaps. Therefore, a dedicated, robust TSL recognition system is crucial for empowering the Tamil-speaking deaf community and improving their communication opportunities.

The Tamil writing system is vast, comprising:

- 12 Uyir Ezuthukkal (Vowels)
- 18 Mei Ezuthukkal (Consonants)
- 216 Uyir Mei Ezuthukkal (Vowel-Consonant combinations)
- 1 Ayudha Ezuthu (Special Character)

1.3 PROBLEM STATEMENT

The central problem addressed by this research is the lack of a robust, scalable, and highly accurate Deep Learning framework for translating Tamil Sign Language (TSL) gestures into audible speech that can handle the full complexity of the Tamil script. Despite existing efforts in Sign Language Recognition (SLR), TSL poses a unique classification challenge due to its requirement for recognizing 248 distinct classes (247 Tamil characters plus a neutral sign) from highly subtle hand gestures. While prior research utilizing the TLFS23 dataset has employed traditional Machine Learning (ML) techniques—such as Random Forest—achieving a high baseline accuracy, these models fundamentally rely on hand-engineered features and lack the ability to

automatically learn the intricate, non-linear spatial dependencies necessary for distinguishing between the 248 classes with sufficient real-world reliability.

Thus, the challenges presented by this research can be grouped into two major areas:

- **Technological Gap:** There is a major technological gap due to the lack of diverse, high-quality datasets for most global sign languages, leading to models that generalize poorly beyond ASL. Current systems also fail to fully capture essential Non-Manual Features like facial expressions, limiting linguistic accuracy. Practical low-power deployment on edge devices remains difficult despite proposed efficient architectures. Moreover, bi-directional translation—especially generating natural 3D signing avatars—remains largely unsolved.
- **Accessibility Gap:** The need to integrate this accurate recognition engine into a seamless, real-time TSL-to-Speech translation system to significantly reduce the social and professional communication barriers faced by the Tamil-speaking deaf and mute community.

1.4 MOTIVATION

The motivation for undertaking this project is rooted in the pursuit of **social equity and enhanced communication accessibility** for a large, underserved population. The ability to communicate effortlessly is a fundamental human right, yet the deaf and mute community in the Tamil-speaking world continues to face significant hurdles that limit their educational and professional growth.

- **Empowerment and Inclusion:** An accurate, real-time TSL translator directly empowers users, offering them independence from human interpreters and allowing for spontaneous, on-demand communication in everyday public and institutional settings. This tool acts as a powerful agent for social inclusion.
- **Maximizing Data Potential:** The existence of the large and comprehensive TLFS23 (**Tamil Language Finger Spelling Image Dataset**) [2] provides an ideal foundation. Our work is motivated by the opportunity to fully leverage this dataset, using the cutting-edge feature-learning capabilities of a

Deep Learning CNN, which is best suited for complex visual pattern classification, to surpass the performance ceilings of prior traditional ML implementations.

- **Technological Advancement:** This project is motivated by a desire to contribute novel, regional-specific research to the global SLR community. By successfully implementing a robust DL framework that efficiently couples MediaPipe's low-latency keypoint extraction with a powerful CNN classifier, we aim to set a new benchmark for TSL recognition, providing a scalable blueprint for similar regional sign language projects worldwide.

1.5 OBJECTIVES

The primary objective of this project is to design, implement, and rigorously evaluate an efficient Deep Learning framework capable of translating static Tamil Sign Language (TSL) finger-spelling gestures into text and speech, thereby significantly enhancing communication accessibility for the Tamil-speaking deaf and mute community.

The specific objectives are defined as follows:

- **To Implement High-Precision Feature Extraction:** Utilize the MediaPipe Hand Landmark Detection framework to accurately and efficiently extract the 21 three-dimensional (x, y, z) hand key points, and process this data into a normalized, coordinate-based feature vector robust to translation and scale variations.
- **To Develop and Optimize a CNN Classifier:** Design and train a custom Convolutional Neural Network (CNN) architecture specifically optimized to process the normalized landmark feature vectors and classify them across all 248 distinct classes of the **TLFS23 dataset**, aiming to achieve classification accuracy exceeding the benchmark set by previous Machine Learning methods.
- **To Integrate an End-to-End Translation Pipeline:** Successfully integrate the CNN classifier output with a Text-to-Speech (TTS) module to provide immediate, audible translation, completing the full TSL-to-Speech communication system.
- **To Validate Real-Time Performance:** Evaluate the computational efficiency, latency (Frames Per Second), and overall robustness of the integrated system in a

simulated real-time environment to ensure its viability for practical deployment on standard computing platforms.

1.6 OVERVIEW OF THE REPORT

The introduction, background, motivation, problem statement, and objectives are explained in **Chapter 1**, highlighting the importance of developing a Tamil Sign Language recognition system.

Existing research on sign language recognition, deep learning models, and MediaPipe-based approaches is reviewed in **Chapter 2**, where the major research gaps relevant to this project are also identified.

The overall system design, including architecture, workflow, and data flow between different components, is described in **Chapter 3**.

The complete methodology—covering dataset preparation, preprocessing, MediaPipe keypoint extraction, and model development using CNN/ResNet50—is detailed in **Chapter 4**.

Outputs from preprocessing, augmentation, landmark detection, and other implementation stages are presented and discussed in **Chapter 5**.

The relevance of this project to the Sustainable Development Goals is highlighted in **Chapter 6**, showing how it supports inclusion, education, innovation, and reduced inequalities.

The overall conclusion of the project and future enhancements—such as real-time webcam integration, Tamil TTS, mobile deployment, and continuous sign recognition—are provided in **Chapter 7**.

CHAPTER 2

LITERATURE SURVEY

The domain of Sign Language Recognition (SLR) is a rapidly evolving field at the intersection of Computer Vision and Deep Learning. This chapter reviews pivotal research that establishes the foundation for, and contributes novel methodologies to, the development of intelligent, real-time sign language translation systems. The literature can be categorized based on the core technology and specific sign language focus.

MAJOR AREAS OF FOCUS:

Researchers emphasize next-generation neural architectures such as Spiking Neural Networks (SNNs) and specialized Active CNN Frameworks to achieve high accuracy while significantly improving energy efficiency for deployment on low-power or edge devices. Studies focus on real-time sequence modeling using recurrent architectures like LSTMs and other intelligent deep learning models to capture the temporal dynamics of continuous signs, crucial for instant text conversion. Literature explores the need for linguistic diversity by presenting solutions for specific, non-Western sign languages, notably Indian Sign Language (ISL) and Thai Sign Language, to broaden global accessibility.

Recent research highlights the delivery of multimodal output, aiming to convert recognized sign gestures into both Text and Speech, ensuring complete communication accessibility between signers and non-signers. Several works propose advanced system integration by investigating the full communication loop, including Sign Language Generation and Avatar Systems, transitioning the field from mere recognition to bi-directional translation and modelling.

2.1 LITERATURE SURVEY

Chaudhari P. et al. (2024) – *Sign Language Recognition using Spiking Neural Networks* [3] This work proposes the use of Spiking Neural Networks (SNNs) for SLR, arguing they provide more energy-efficient computations and can process temporal sequences with simpler architectures than conventional Artificial Neural Networks (ANNs). The SNN achieved an improvement in accuracy (+2.70%) over the previous state-of-the-art when evaluated on the WLASL300dataset. A core finding is that the SNN had significantly sparser activations (25% less), which relates to a lower power requirement for deployment. This demonstrates the potential for implementing highly effective and efficient SLR systems on edge devices.

Rastgoo et al. (2021) – *Sign Language Recognition: A Deep Survey* [4] This work provides a comprehensive taxonomy of the SLR field, surveying deep learning methods for both isolated and continuous signing. The authors discuss key challenges such as data scarcity and the critical need for signer-independent recognition, classifying existing approaches (vision-based vs. sensor-based) to establish the state-of-the-art and outline the major research gaps.

"Conversion of Sign Language to Text Using Machine Learning" (2023) – *Springer Professional* [5] This chapter introduces a customized Convolutional Neural Network (CNN) model designed for recognizing sign language gestures. Focusing on accurate feature extraction, the model demonstrates a fundamental, vision-based approach to converting gestures to text. The authors emphasize that CNNs offer a practical solution by reducing the complexity of pre-processing steps.

Toshpulatov et al. (2025) – *Deep Learning Pathways for Automatic Sign Language Processing* [6] This paper investigates advanced deep learning architectures, including Transformer models, that form the backbone of modern SLR. The focus is on overcoming challenges in Continuous Sign Language Recognition (CSLR), such as the weak supervision of sign labels, which is critical for handling the long-term temporal and contextual complexity of translating signed sentences.

Gao et al. (2020) – *Sign Language Recognition with Deep Learning: A Survey* [7] This is a highly comprehensive and foundational survey that focuses entirely on deep learning methods for Sign Language Recognition. It categorizes approaches, discusses key datasets, and provides a clear breakdown of how CNNs, RNNs (LSTMs), and other architectures are applied

ZainEldin et al. (2024) – *Active Convolutional Neural Networks Sign Language Framework (ActiveCNN-SL)* [8] This work proposes a specialized framework for improving SLR performance through model optimization. The ActiveCNN-SL likely integrates active learning or optimization techniques within the CNN structure to enhance feature extraction and boost recognition accuracy, offering a path for designing highly performant and targeted gesture recognition systems.

Tripathi et al. (2025) – *Intelligent Sign Language Recognition for Real-Time Text* [9] This research directly addresses the demand for practical, instantaneous translation by presenting an optimized model for Intelligent Sign Language Recognition for Real-Time Text. The focus is on robust deep learning architectures, often incorporating optimization algorithms, to ensure the system is fast, reliable, and deployable for real-world scenarios requiring immediate text output.

Ingoley et al. (2025) – *Interpretation of Indian Sign Language to Text and Speech* [10] This paper addresses the vital need for linguistic diversity in SLR by focusing on the interpretation of Indian Sign Language (ISL). The system aims to convert recognized ISL signs from live video into both Text and Speech, providing a necessary multimodal output solution to bridge the communication gap effectively between signers and non-signers.

Jintanachaiwat et al. (2024) – *Using LSTM to Translate Thai Sign Language to Text in Real-Time* [11] This study validates the effectiveness of the Long Short-Term Memory (LSTM) architecture for sequential translation tasks. By combining LSTMs with pose estimation techniques (like MediaPipe Holistic), the

researchers successfully translated Thai Sign Language to Text in Real-Time, demonstrating the capability of recurrent networks to handle the temporal complexities of continuous, dynamic signing.

Wolfe et al. (2025) – *Special Issue on Sign Language Translation and Avatar Systems* [12] This special issue signals the field's current high-level research ambition. It highlights the shift in research priority from isolated recognition to complex, sentence-level Sign Language Translation and the integration of Avatar Systems for production. This work defines the future scope of the project, focusing on complete, bi-directional communication systems.

2.2 RESEARCH GAP

2.2.1 LACK OF RESOURCE AVAILABILITY FOR GLOBAL LINGUISTIC DIVERSITY

Research is heavily biased toward American Sign Language (ASL) due to dataset availability. There is a severe scarcity of large, high-quality, annotated video datasets for most other sign languages, including structurally distinct variants like Indian Sign Language (ISL) and Thai Sign Language. This resource gap prevents the development of globally applicable SLR systems (Ingoley et al., 2025; Jintanachaiwat et al., 2024).

2.2.2 LACK OF NON-MANUAL FEATURE INTEGRATION

Current deep learning systems often treat signs as purely manual gestures, overlooking the grammatical and semantic role of Non-Manual Features (NMFs)—such as facial expressions, eyebrow movements, and mouth shapes. The inability to fully and accurately integrate these NMFs results in systems that are linguistically incomplete and prone to misinterpreting complex signs or sentence structures.

2.2.3 PRACTICAL CHALLENGES IN ACHIEVING TRUE EDGE-DEVICE EFFICIENCY

While highly efficient architectures like Spiking Neural Networks (SNNs) are proposed for low-power edge deployment (Chaudhari P. et al., 2024), achieving the required levels of sparsity and minimizing computational overheads (e.g., memory

access, data movement) in practice remains difficult. There is a gap between theoretical energy savings and real-world deployment where factors like latency and power consumption must be optimized simultaneously without significant accuracy loss.

2.2.4 UNSOLVED PROBLEM OF HIGH-FIDELITY BI-DIRECTIONAL TRANSLATION

The field is focused on Recognition (Sign $\rightarrow$ Text), but there is a lack of mature, integrated systems for Sign Language Production (SLP). Creating socially acceptable, emotionally accurate, and linguistically precise 3D Avatars capable of continuous, natural signing from text input is a challenge that defines the future of the field, requiring significant advancements in generation and modeling technologies (Wolfe et al., 2025).

CHAPTER 3

SYSTEM DESIGN

3.1 PROCESS FLOW

The system starts by organizing Tamil Sign Language images into separate class folders, creating a structured dataset for model training. Each image is then passed through a pre-processing pipeline, where it is loaded, resized to a uniform dimension, normalized to stabilize pixel values, and augmented using transformations such as rotation, flipping, zooming, and brightness adjustments to increase dataset variety.

After pre-processing, the images are processed through the feature extraction stage. Here, a CNN or ResNet50 model automatically learns meaningful gesture features, including hand shapes, edges, and finger positions. These extracted features are then sent to the classification stage, where fully connected layers analyze them and assign the appropriate Tamil sign label.

During training, the model continuously learns from the processed data, improving its accuracy and minimizing loss over multiple epochs. When a new image or hand-sign input is provided, it is passed through the same pre-processing and feature extraction steps, and the trained model generates the corresponding Tamil text output. This workflow represents the complete pipeline of the Tamil Sign Language to Text Conversion System

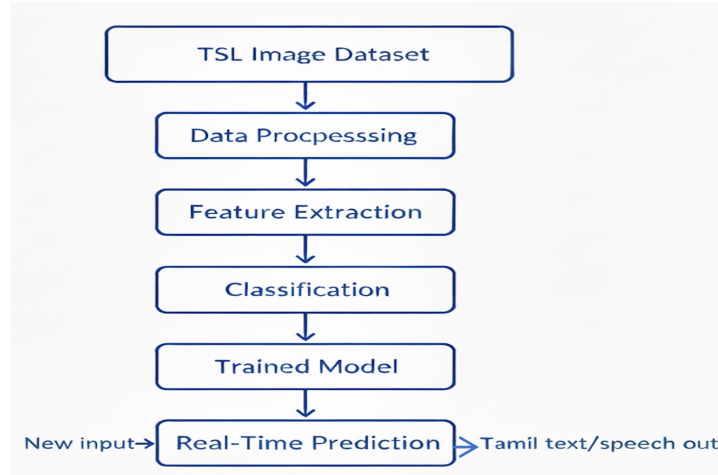


Figure 2: Flow Diagram

3.2 ARCHITECTURE

The architecture of the Tamil Sign Language to Text Conversion System consists of five major components: Data Pre-processing, Feature Extraction, Deep Learning Model Training, Prediction Pipeline, and the User Interface Layer. These components work together to convert static images and real-time webcam input into meaningful Tamil text.

3.2.1 DATA PREPROCESSING LAYER

The system begins with a pre-collected dataset containing images of Tamil Sign Language gestures. These images are loaded and automatically labeled based on their respective class folders. Each image is then resized to a fixed dimension to ensure consistency across the dataset. The images are normalized to convert raw pixel values into a standard numeric range, improving the stability of model training. To enhance dataset variability and prevent overfitting, augmentation techniques such as rotation, flipping, zooming, and brightness adjustments are applied.

3.2.2. FEATURE EXTRACTION LAYER (MEDIAPIPE)

After pre-processing, each image is passed to the MediaPipe framework for hand detection. MediaPipe identifies the region containing the hand and extracts 21 landmark points representing finger joints, wrist position, and structural orientation. These landmarks are processed to form a numerical feature vector that captures the

spatial relationships of the gesture. This feature vector forms the input to the deep-learning model.

3.2.3 DEEP LEARNING COMPONENT (CNN ARCHITECTURE)

The feature vector is fed into the CNN architecture during training. The model consists of several internal layers

- **Convolutional Layers** that learn spatial patterns from the hand landmarks
- **ReLU Activation** to introduce non-linearity
- **Pooling Layers** to reduce spatial dimensions
- **Flatten Layer** to convert matrix data into a vector
- **Dense Layer** for class pattern learning
- **Softmax Output Layer** that generates probability scores for all 248 Tamil sign classes

Once trained, this CNN becomes the **Trained CNN Model**, which is used during real-time prediction.

3.2.4. PREDICTION LAYER

During real-time usage, frames from a webcam are continuously captured and processed. Each frame undergoes feature extraction using MediaPipe, which identifies and extracts hand landmark points. These features are passed to the CNN Interface, which uses the trained CNN model to classify the gesture. The predicted class is converted into Tamil text, representing the recognized sign.

3.2.5. UI LAYER

The final predicted text is displayed to the user through the User Interface Layer. This UI presents the recognized Tamil character or word, enabling smooth interaction between the signer and the system.

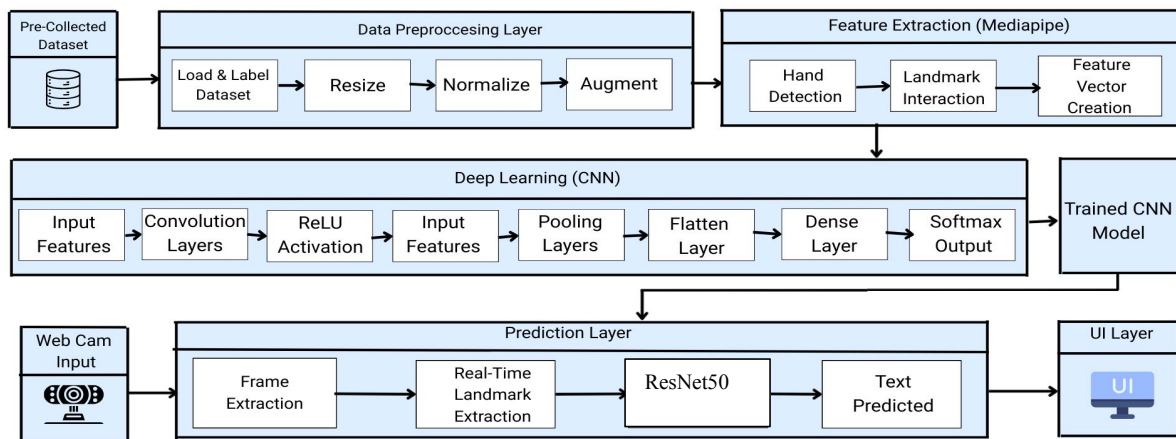


Figure 3: System Architecture

CHAPTER 4

METHODOLOGY

The proposed system is designed to automatically recognize hand signs and convert them into text. The methodology involves five major stages: dataset acquisition, pre-processing using Open CV and Media Pipe, data augmentation, feature extraction using the ResNet50 deep learning model, and classification using a CNN-based architecture. The workflow ensures that input images undergo consistent pre-processing before being fed into a robust deep learning model capable of extracting high-level features for accurate gesture classification.

4.1 DATASET PREPARATION

The dataset contains images of hand signs captured under varying lighting conditions, backgrounds, and orientations, ensuring real-world diversity and robustness for Tamil sign recognition. These images are organized into separate folders, with each folder representing a specific Tamil hand sign class, enabling automatic labeling and structured class mapping. To load the dataset efficiently, TensorFlow's `image_dataset_from_directory()` function is used, which reads images directly from the directory structure, assigns labels based on folder names, and resizes all images to a uniform input size of 224×224 pixels, suitable for CNN architectures like ResNet50. This pre-processing step ensures standardization, maintains batch consistency, and prepares the dataset in a format appropriate for deep learning pipelines. Additionally, shuffling is applied to avoid learning order-based biases, while optional augmentation techniques such as rotation, flipping, and zooming further enhance model generalization. The prepared dataset is later passed into the CNN training stage, enabling the model to learn discriminative spatial and visual features required for accurate real-time Tamil hand sign prediction.

4.2 PREPROCESSING

Preprocessing is a vital stage in the proposed hand-sign recognition system, ensuring that all images are standardized before being used to train the deep-learning model. As shown in the pseudocode, pre-processing begins by defining the dataset path, image size, and batch size, followed by loading images using the `image_dataset_from_directory()` function, which automatically assigns labels based on

folder names and resizes every sample to 224×224 pixels to match CNN requirements. The dataset is then optimized for efficient training through caching, shuffling, and prefetching—where caching stores processed images in memory for faster access, shuffling randomizes input order to prevent pattern learning, and prefetching overlaps data loading with model execution to improve throughput. A normalization layer ($\text{Rescaling}(1./255)$) is then applied to convert pixel values from 0–255 into a 0–1 range, improving gradient stability and model convergence. To verify pre-processing correctness, five random images are extracted, passed through the normalization layer, and displayed along with their class labels. This structured pre-processing pipeline ensures that all images entering the CNN have consistent dimensions, pixel distribution, and format, ultimately improving training quality and recognition accuracy.

Pseudocode:

- SET `dataset_path`, `image_size`, `batch_size`
- LOAD dataset using `image_dataset_from_directory()`
- APPLY `cache()`, `shuffle()`, and `prefetch()` for performance
- DEFINE `preprocess_layer = Rescaling(1/255)`
- RANDOMLY SELECT 5 images from dataset
- APPLY `preprocess_layer` to selected images
- DISPLAY processed images with labels

4.3 ALGORITHM – ResNet50

ResNet50 is a deep Convolutional Neural Network consisting of 50 trainable layers, designed to extract highly discriminative visual features from images. In this project, ResNet50 is used to learn the unique hand shapes, finger positions, and structural variations present in Tamil Sign Language gestures. The architecture begins with an input layer, where each preprocessed and augmented image of size $224 \times 224 \times 3$ enters the network. This is followed by a 7×7 convolution layer and a 3×3 max-pooling operation, which help capture low-level visual patterns such as edges, curves, and contours of the hand.

The core body of ResNet50 contains multiple residual blocks, each made up of three convolution layers — a 1×1 convolution for dimensionality reduction, a 3×3 convolution for spatial feature extraction, and another 1×1 convolution for restoring channel depth. These blocks are grouped into four stages, increasing in depth and complexity. The key characteristic of ResNet50 is the skip (or shortcut) connection, where the input of a residual block is added directly to its output. This prevents vanishing gradients and enables effective training even with large numbers of layers. For gesture recognition, this mechanism ensures that fine-grained finger movements and subtle variations in hand posture are preserved throughout the network.

As images progress deeper into the architecture, the network transforms them into high-level feature representations, allowing the model to distinguish between similar

Tamil signs. After passing through all residual blocks, the feature maps undergo 7×7 global average pooling, which compresses spatial information into a compact feature vector. Finally, a fully connected (FC) layer with softmax activation generates class probabilities, predicting which Tamil sign the input image represents.

Formula of ResNet50 (Residual Block)

The fundamental equation of a residual block is:

$$y = F(x) + x$$

Where:

- x = input to the block
- $F(x)$ = residual function learned by stacked convolution layers
- y = output of the block

This means the network learns:

$$F(x) = H(x) - x$$

Where $H(x)$ is the desired mapping.

This reduces learning complexity and stabilizes gradient flow.

Why ResNet50 Suits This Project

It is capable of learning complex hand-gesture patterns, finger positions, and subtle visual variations.

It efficiently handles large datasets with a high number of classes, such as 248

Tamil Sign Language characters.

Its skip-connection (residual) architecture prevents vanishing gradients and performance degradation during deep training.

Pretrained ImageNet weights provide a strong starting point, improving accuracy and reducing training time.

It extracts high-level semantic and spatial features required for accurate Tamil sign recognition.

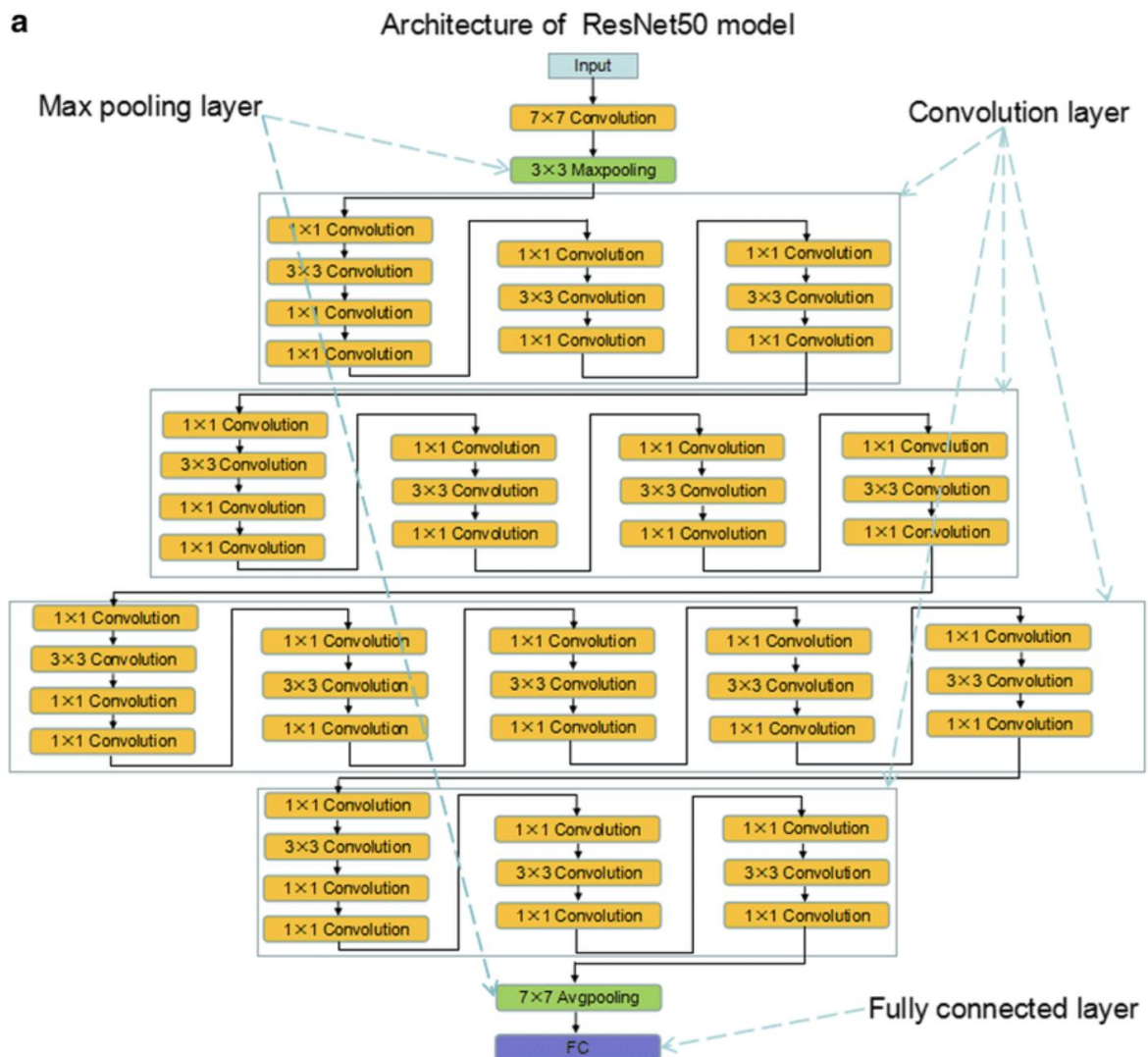


Figure 4: Architecture diagram of ResNet50[13]

4.4 MEDIAPIPE DEFINITION

MediaPipe Hands is used in this project to detect, track, and represent the spatial structure of the human hand using 21 predefined landmark points. The diagram illustrates how each landmark corresponds to a specific anatomical location, including the wrist, finger joints, and fingertips. Landmark 0 represents the wrist, while landmarks 1 to 4 correspond to the thumb, 5 to 8 represent the index finger, 9 to 12 represent the middle finger, 13 to 16 represent the ring finger, and 17 to 20 represent the little finger (pinky). These coordinates are extracted in 3D space as (x, y, z) values, where x and y indicate pixel positions on the image, and z provides depth information relative to the wrist.

This structured landmark representation captures the geometric pose of the hand, including finger direction, angle, curvature, and relative distances, which are essential in differentiating Tamil Sign Language gestures. By focusing only on the detected hand landmarks rather than the entire image, MediaPipe effectively removes background noise, lighting variations, and unnecessary objects. MediaPipe also performs temporal smoothing to reduce jitter and ensure stable tracking during hand movement. The extracted landmark features are then passed to the deep-learning classification model, improving accuracy, consistency, and computational efficiency in sign recognition.

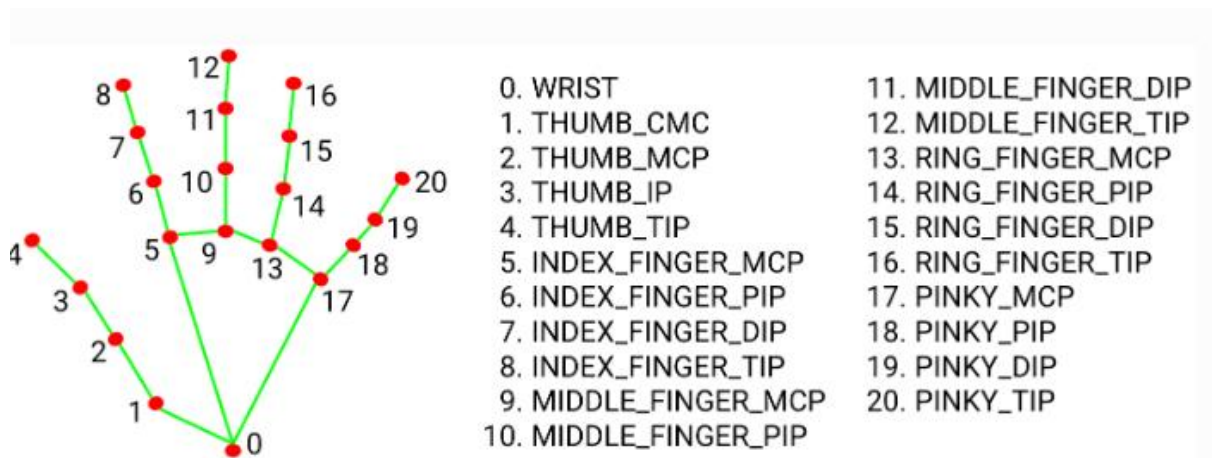


Figure 4: hand landmarks [14]

The **MediaPipe architecture** shown here illustrates the complete real-time hand-tracking pipeline, where each frame from the camera flows through a sequence of optimized processing stages to achieve fast and accurate landmark detection. The pipeline begins with a RealTimeFlowLimiter that stabilizes the incoming frames, after which hand detection is triggered only on the first frame or when the hand is not found. Detected regions are converted into rectangular bounding boxes and then passed to an ImageCropping module that extracts only the hand area. This cropped image is then processed by the HandLandmark model to generate 21 key landmark points along with a hand-presence flag. These landmarks are further converted back into tracking rectangles for use in the next frame, while a separate rendering step generates the annotated landmark image for display. The architecture includes a temporal feedback loop (blue edges) to reuse previous-frame detections, significantly improving speed and supporting smooth real-time performance.

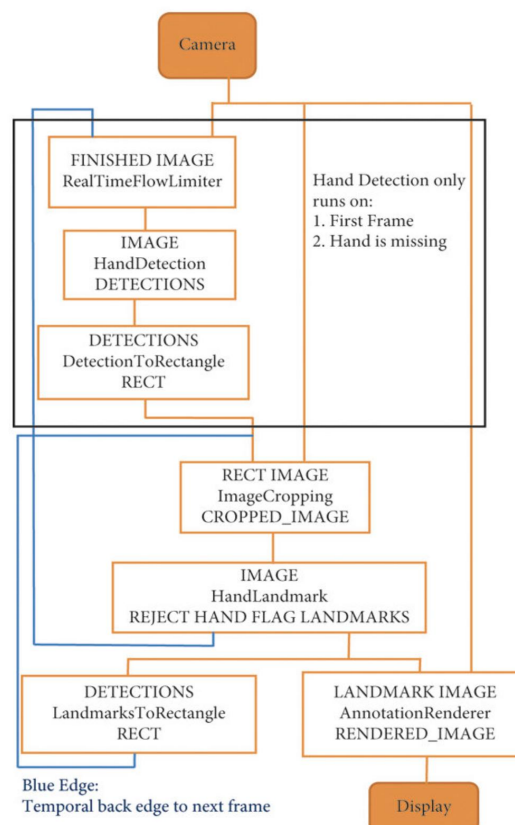


Figure 5:MediaPipe Architecture

CHAPTER 5

RESULTS AND DISCUSSION

5.1 DATASET DESCRIPTION

The dataset used in this project is the Tamil Sign Language Hand Gesture Dataset, consisting of multiple hand-sign images representing different Tamil characters and gestures. The dataset is organized into class-wise folders, where each folder corresponds to a specific Tamil sign. In total, the dataset contains 248 classes, with at least 1,000 images per class, contributing to a massive 2,55,155 images overall. Each class represents a Tamil character finger-spelled by participants belonging to diverse age groups, ensuring natural variation in hand shape, size, and posture. The dataset covers the full Tamil character set, including 12 Uyir ezuthukal (vowels), 18 Mei ezuthukal (consonants), 216 Uyir-Mei ezuthukal (vowel-consonant combinations), and 1 Ayudha ezuthu (special character), providing unique binary and visual representations for all 247 Tamil characters, along with an additional gesture class.

All images were collected under varying background conditions, lighting environments, hand orientations, and camera distances to ensure robustness and real-world applicability. During preprocessing, every image was resized to 224×224 pixels, and class labels were automatically generated based on folder names. To further improve dataset diversity and avoid overfitting, data augmentation techniques such as horizontal flipping, random rotation, zooming, contrast adjustments, and brightness variations were applied. These augmentations enabled the model to generalize better and perform reliably, even when the gesture appears under different orientations, lighting conditions, or environmental settings.



Figure 6: Samplimages[15]

5.2 SOFTWARE SPECIFICATION

The software specifications outline the essential tools and frameworks required to build and execute the Tamil Sign Language to Text Conversion system. These components ensure effective model development, hand landmark detection, and smooth experimentation. The table below summarizes the software environment used in the project.

Table 1:Software Specifications

Software Component	Version
Programming Language	Python 3.8 / 3.9 / 3.10
Deep Learning Framework	TensorFlow
Hand Landmark Detection	MediaPipe Hands Framework
Development Environment	Jupyter Notebook / VS Code
Operating System	Windows 10 / Windows 11

- Python 3.x is used as the main programming language because it is simple, readable, and supports powerful machine learning and computer vision libraries.
- TensorFlow is required for building, training, and evaluating the deep-learning model, offering neural network APIs, GPU acceleration, and efficient optimization.
- MediaPipe Hands Framework is utilized to detect and extract hand landmarks, helping isolate gesture information accurately before classification.
- Jupyter Notebook and Visual Studio Code serve as development environments, enabling interactive coding, visualization, debugging, and iterative experimentation.
- Windows 10 / Windows 11 is chosen as the operating system due to its compatibility with required libraries, drivers, and hardware configurations.

- Together, these software tools provide a stable, efficient, and reliable platform for developing and validating the Tamil Sign Language to Text Conversion system.

5.3 HARDWARE SPECIFICATION

The hardware specifications define the computational setup necessary for efficient data processing, model training, and real-time prediction. These requirements ensure optimal system performance without bottlenecks. The table below lists the hardware configuration used for the project.

Table 2:Hardware Specifications

Hardware Component	Specification
Processor (CPU)	Intel Core i5 / i7 or equivalent
RAM	16 GB (Recommended)
GPU	NVIDIA 3060 12GB VRAM (Recommended)
Storage	Minimum 10 GB Free Storage
System Type	64-bit Operating System

- Intel Core i5 / i7 processor or equivalent CPU is required to efficiently handle dataset loading, preprocessing, and general computational tasks without slowing down.
- Minimum 16 GB RAM is recommended to support large datasets, real-time image processing, and deep-learning operations without memory bottlenecks.
- NVIDIA GPU (e.g., RTX 3060 with 12 GB VRAM) is suggested to accelerate TensorFlow-based model training, as CUDA-enabled GPUs significantly reduce computation time.
- At least 10 GB of free storage is necessary for storing the dataset, trained model files, preprocessing outputs, and required libraries.
- 64-bit operating system ensures compatibility with machine learning frameworks, GPU drivers, and high-performance computing tools.

- Together, these hardware components provide a stable, optimized, and high-performance environment for developing, testing, and deploying the Tamil Sign Language recognition system.

5.4 EXPERIMENTATION

In the current stage of the project, the implementation has successfully covered dataset preparation, preprocessing, and data augmentation. The dataset containing hand-sign images was loaded and categorized using TensorFlow's `image_dataset_from_directory()` method, which automatically assigned class labels based on folder names and resized all images to 224×224 pixels. This ensured that the dataset maintained uniformity in input dimensions, which is essential for further processing.

The preprocessing of images was carried out using a rescaling operation that normalized pixel values from the range 0–255 to 0–1. This was implemented using the TensorFlow Rescaling ($1./255$) layer. Normalization plays a major role in stabilizing gradient flow and improving model learning efficiency. To validate preprocessing, a set of five random images from the dataset were visualized after normalization. The visual inspection confirmed that all images were correctly normalized and prepared for input into the deep-learning model.

To enhance the dataset further and improve its diversity, five types of data augmentation techniques were applied: random horizontal flipping, rotation, zooming, contrast adjustment, and brightness adjustment. These augmentations helped simulate different real-world variations such as hand orientation changes, lighting differences, and camera distance variations. Augmented samples were also visualized, confirming that the transformations were correctly applied and that the augmented images preserved the structure of the original hand gesture.

5.5 RESULT

The outcomes of the current stage of the project demonstrate successful preparation of the Tamil Sign Language dataset and readiness for subsequent model development. A large, diverse dataset consisting of 248 gesture classes and more than 2,55,000 images was successfully collected, organized, and recognized by the system.

Each class was correctly mapped to its corresponding Tamil character, confirming accurate dataset structuring and label assignment. All images were uniformly resized to 224×224 pixels, resulting in a standardized dataset compatible with deep-learning architectures such as CNN and ResNet50.

The preprocessing pipeline produced consistent and reliable results, converting all raw images into normalized numerical form suitable for efficient neural-network learning. Visual inspection confirmed that normalization preserved gesture clarity, hand structure, and background integrity, indicating no loss of essential feature information. The system demonstrated the ability to process large-scale image batches smoothly, validating the effectiveness of the data pipeline and confirming that the hardware–software configuration adequately supports high-volume computational tasks.

To further strengthen dataset quality, augmentation outputs successfully expanded visual diversity while retaining gesture meaning. The augmented images reflected realistic variations in rotation, zoom, contrast, orientation, and lighting, proving that the system can generate additional training samples capable of improving model generalization. These results verify that the dataset is now more robust and capable of representing real-world user conditions.



Figure 7: Input image

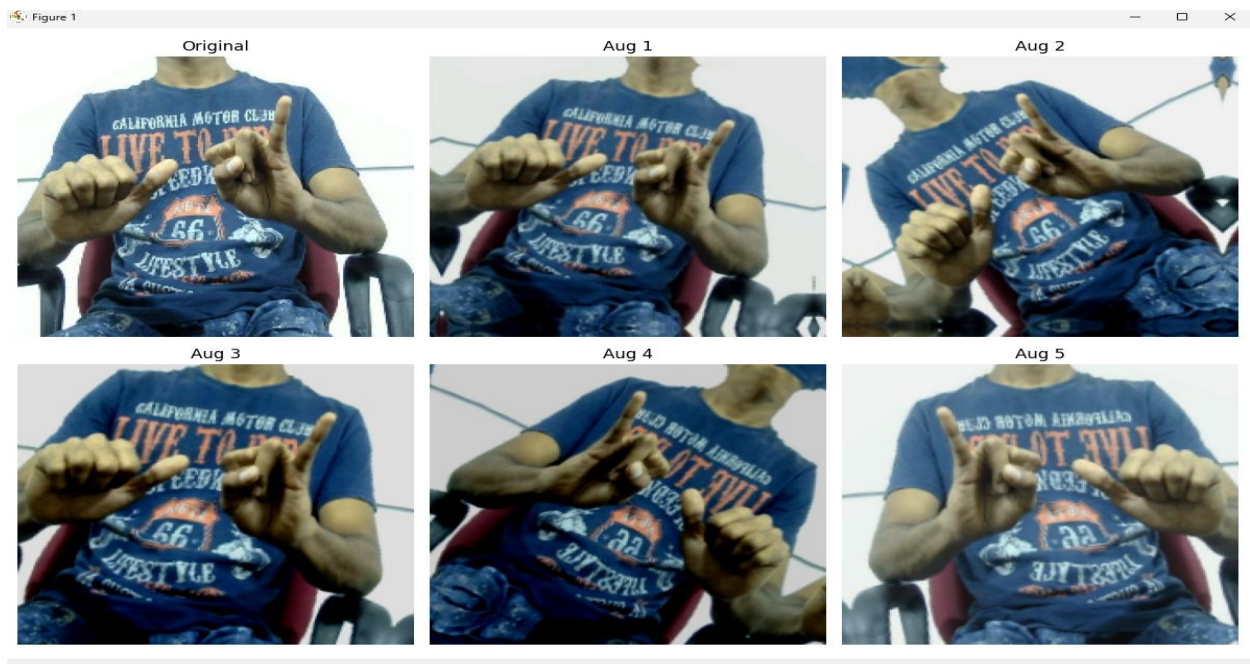


Figure 8: Output image

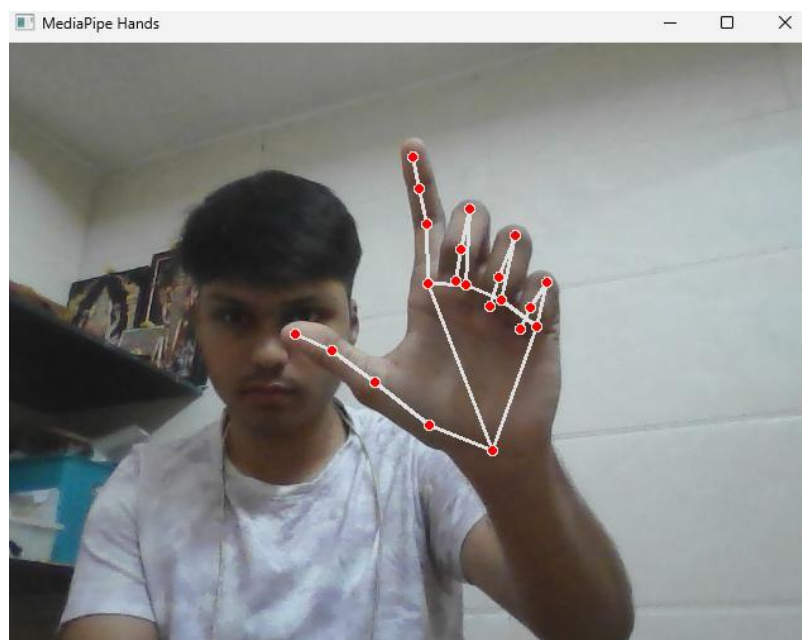


Figure 9: Landmark point

CHAPTER 6

SUSTAINABLE DEVELOPMENT GOALS

6.1 CONTRIBUTION TO SDGs

The development of a Tamil Sign Language to Text Conversion system contributes significantly to several United Nations Sustainable Development Goals (SDGs) by addressing accessibility, education, innovation, and inclusivity in communication technologies. This project is aimed at empowering individuals who rely on Tamil Sign Language for daily interaction, particularly those who are hearing or speech-impaired, and enhancing their ability to communicate effectively with the broader Tamil-speaking community.

- **SDG 10: Reduced Inequalities**

Helps break communication barriers faced by hearing and speech-impaired individuals. Converts Tamil Sign Language gestures into readable Tamil text, enhancing accessibility. Encourages equal participation in education, public services, and social activities. Promotes dignity, confidence, and independence among differently-abled individuals. Supports disability-inclusive policies and community engagement within Tamil-speaking regions.

- **SDG 4: Quality Education**

Acts as a digital learning tool for students, teachers, and caregivers to understand Tamil Sign Language. Can be integrated into inclusive classrooms and special education programs. Encourages awareness and acceptance of sign language in mainstream education. Supports self-learning platforms, e-learning apps, and remote education. Helps bridge communication gaps between educators and hearing-impaired learners.

- **SDG 9: Industry, Innovation, and Infrastructure**

Demonstrates practical use of AI, CNNs, MediaPipe, and computer vision in assistive technology. Encourages development of innovative accessibility-based digital infrastructure. Lays the foundation for automated sign-

language translation systems in public and private sectors. Can be extended into mobile apps, kiosks, customer service systems, and healthcare communication tools. Inspires further research and technological advancement in gesture recognition.

- **SDG 8: Decent Work and Economic Growth**

Enhances employability of hearing-impaired individuals by improving communication in workplaces. Supports inclusive recruitment, training, teamwork, and career advancement. Reduces dependency on interpreters, enabling self-reliance in professional settings. Encourages companies to adopt accessibility technologies and inclusive hiring practices.

Helps create equal economic opportunities and workplace diversity.

- **SDG 3: Good Health and Well-being**

Improves mental well-being by reducing social isolation and communication stress. Can support communication in hospitals, clinics, and counseling centers. Encourages emotionally supportive interaction between patients and healthcare providers.

- **SDG 5: Gender Equality**

Provides equal access to communication tools for women and girls with hearing impairments. Helps remove social and educational barriers for marginalized female sign-language users.

- **SDG 17: Partnerships for the Goals**

Enables collaboration between academic institutions, NGOs, disability organizations, and government bodies. Encourages cross-disciplinary research linking AI, linguistics, healthcare, and social welfare.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

In this project, the initial stages of the Tamil Sign Language to Text Conversion system have been successfully completed. The dataset was loaded and organized properly using TensorFlow, and all images were resized to a uniform dimension to ensure consistency. Preprocessing was implemented using normalization, which converted pixel values into a suitable range for model training. A set of random preprocessed images was also visualized to confirm that the input pipeline was working correctly.

In addition, five different data augmentation techniques horizontal flipping, rotation, zooming, contrast adjustment, and brightness variation were applied to improve dataset diversity and model robustness. Augmented samples were successfully displayed to verify that the transformations were applied as expected. Overall, the dataset preparation, preprocessing, and augmentation stages have been completed, forming a strong foundation for the next steps of feature extraction and model training.

7.2 FUTURE WORK

Although the current work has successfully completed dataset preparation, preprocessing, and data augmentation, there are several important enhancements and extensions planned for the next stages of the project. The immediate future work includes implementing and training the deep-learning model using CNN or ResNet50 and evaluating its accuracy using standard metrics. Once the model achieves satisfactory performance, real-time gesture detection using a live webcam will be introduced to make the system interactive and practical for everyday use.

The future scope also includes expanding the dataset by adding more Tamil sign symbols, dynamic gestures, and various hand orientations to improve the system's scalability. Converting the recognized Tamil text into speech using a Tamil Text-to-Speech (TTS) engine will enable two-way communication between sign language users and non-sign language speakers. Another planned enhancement is the

development of a mobile application using TensorFlow Lite, allowing users to perform sign recognition on their smartphones without requiring high-end hardware. Further improvements can include integrating a bidirectional translation system where not only signs are converted to Tamil text but also Tamil text is converted into animated Tamil sign gestures using 2D/3D avatars.

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APENDEX

PAPER PUBLICATION



4th International Conference on Automation, Signal Processing, Instrumentation and Control : Submission (21) has been created.

1 message

Microsoft CMT <noreply@msr-cmt.org>
To: 221801035@rajalakshmi.edu.in

Hello,

The following submission has been created.

Track Name: iCASIC2026

Paper ID: 21

Paper Title: A Deep Learning Framework for Translating Tamil Sign Language Gestures into Speech

Abstract:

Communication is essential in the interaction of people, but the hearing or speech impaired people are normally hindered in communicating with other people who cannot understand sign language. To overcome this communication barrier, the suggested system proposes a model of Tamil Sign Language Recognition with the help of Convolutional neural networks (CNNs) combined with Mediapipe to track the hands and recognize gestures in real-time. It is a system that records live video with a webcam and isolates 21 hand landmarks with Mediapipe hand detection module. The distance features in the form of numbers between the spatial relations of these landmarks are transformed into numerical data, which are utilized to train a CNN model that can identify and categorize various Tamil gestures. The model is trained using gesture datasets acquired on various Tamil letters or words and shows to be able to recognize efficiently, using deep learning. In real-time prediction, the system detects the gesture, recognizes it with the help of the trained CNN model, and shows the resulting Tamil letter or word on the screen by the means of Unicode font displayed. The technology is a societal solution to communication and support that is fast, precise, and accessible to users and leads to an inclusive digital environment in the hearing and speech-impaired population.

Created on: Wed, 12 Nov 2025 07:30:27 GMT

Last Modified: Wed, 12 Nov 2025 07:30:27 GMT

Authors:

- 221801035@rajalakshmi.edu.in (Primary)
- 221801036@rajalakshmi.edu.in
- 221801053@rajalakshmi.edu.in
- beulah.a@rajalakshmi.edu.in
- sureshkumar.s@rajalakshmi.edu.in

Secondary Subject Areas: Not Entered

Submission Files:

A Deep Learning Framework for Translating Tamil Sign Language Gestures into Speech.docx (176 Kb, Wed, 12 Nov 2025 07:25:30 GMT)

Submission Questions Response: Not Entered

Thanks,
CMT team.

Fig a1: Paper Confirmation Mail

A Deep Learning Framework for Translating Tamil Sign Language Gestures into Speech

Suresh Kumar S
Professor of Artificial Intelligence and
Data science
Rajalakshmi Engineering College
Chennai, India
sureshkumar.s@rajalakshmi.edu.in

Beulah A
Assistant Professor of Artificial
Intelligence and Data science
Rajalakshmi Engineering College
Chennai, India
Beulah.a@rajalakshmi.edu.in

Nikilesh S
Artificial Intelligence and Data science
Rajalakshmi Engineering College
Chennai, India
221801035@rajalakshmi.edu.in

Poongavin B
Artificial Intelligence and Data science
Rajalakshmi Engineering College
Chennai, India
221801036@rajalakshmi.edu.in

Sukish M
Artificial Intelligence and Data science
Rajalakshmi Engineering College
Chennai, India
221801053@rajalakshmi.edu.in

Abstract—Communication is essential in the interaction of people, but the hearing or speech impaired people are normally hindered in communicating with other people who cannot understand sign language. To overcome this communication barrier, the suggested system proposes a model of Tamil Sign Language Recognition with the help of Convolutional neural networks (CNNs) combined with Mediapipe to track the hands and recognize gestures in real-time. It is a system that records live video with a webcam and isolates 21 hand landmarks with Mediapipe hand detection module. The distance features in the form of numbers between the spatial relations of these landmarks are transformed into numerical data, which are utilized to train a CNN model that can identify and categorize various Tamil gestures. The model is trained using gesture datasets acquired on various Tamil letters or words and shows to be able to recognize efficiently, using deep learning. In real-time prediction, the system detects the gesture, recognizes it with the help of the trained CNN model, and shows the resulting Tamil letter or word on the screen by the means of Unicode font displayed. The technology is a societal solution to communication and support that is fast, precise, and accessible to users and leads to an inclusive digital environment in the hearing and speech-impaired population.

Keywords—Tamil Sign Language (TSL), Sign Language Recognition (SLR), MediaPipe, Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Deep Learning, Hand Gesture Recognition, Real-Time System, Accessibility.

I. INTRODUCTION

Communication is critical in helping people to interact with one another but people with hearing or speech difficulties usually encounter a lot of challenges when communicating with other people who may not comprehend the sign language. This challenge has been made worse by the fact that there is no universal interpretation of sign language something that in many instances forces one to rely on human interpreters that are not always viable or accessible. Also, the majority of the available sign language systems are based on the English language or any other international language, which leaves Tamil Sign Language (TSL) with very little technological support.

Therefore, it is evident that the automated system that will be capable of properly identifying Tamil hand gestures and converting them into readable text is desired. This project is specifically done to address the communication gap between the hearing impaired and

Tamil speaking community and will facilitate the inclusion of communication by this community. Through creation of real-time system, it is aimed that communication among the differently-abled individuals will be easier and more self-reliant since the human interpreters are not always feasible and accessible.

To attain this, the suggested system presents a Tamil Sign Language Recognition model based on the Convolutional Neural Networks (CNNs) with Mediapipe to track and identify gestures in real-time. This system uses a webcam to capture video in real-time and uses Mediapipe to identify 21 distinct hand landmarks in this image, which is then transformed into the set of distance features that can be digitized. The CNN model is then trained with these powerful features, and it is programmed to recognize the various Tamil gestures correctly thereby allowing real time recognition of Tamil gestures.

The main goals include creating an automated system to recognize TSL gestures with the help of deep learning (CNN) and the ability to translate Tamil gestures into text in real-time to make them easier to access. The last system is quick, precise and convenient system which predicts the gesture and appears on the screen with the respective Tamil letter or word instantly. Finally, this work will work towards inclusive online accessibility, which will facilitate autonomy and assist the hearing and speech-impaired community communicate better in any social setting.

II. RELATED WORKS

The Basic dilemma Isabelle Isabelle and the Imperative of History: The basic driving force behind the decades of research on Sign Language Recognition (SLR) is the fact that there is the necessity to lessen the seriousness of communication barriers the world deaf-mute community is experiencing. The dependence on human translators cannot be expanded and omnipresent which means that the systems which will enable the lack of reliance and inclusiveness in daily life and workplaces should be developed.

First Sensor Based Systems and their flaws:

The initial years of the SLR research were more inclined towards sensor based technologies. They were novel devices, which relied on specialized hardware, such as computerized data gloves, whether wired or wireless, and large depth sensing cameras (e.g., Microsoft Kinetic). Though these methods might provide high-quality kinematic data on the movements of fingers and joints, their high cost, complex calibration requirements and invasive qualities

rendered them in practice generally, which in a sense limited their possible contribution to universal access.

The Marker-less SLR to Vision-based Shift:

It was observed that the specialized hardware had its disadvantages, and thus there was a shift towards critical paradigm shift to vision-based SLR which was made out of the available, off-the-shelf, RGB cameras that were prevalent in most modern devices. The first strategies towards the vision would involve the normal computer vision algorithms like the background subtraction algorithms, color-based segmentation algorithms, and hand-trained features (e.g., SIFT, HOG) to locate and describe the hand. Such systems however were unable to cope with real world complexity, e.g., in different lighting conditions, occlusions and cluttered backgrounds, to offer consistent performance in unconstrained environments.

A Deep Learning Revolution in feature Extraction:

Deep learning (DL) and, more specifically, Convolutional Neural Networks (CNNs) introduced a new breakthrough into the sphere since the models can now extract meaningful features on raw image data without being limited by the hand-crafted features. This has altered the focus of isolating the entire hand image in a dissimilar and accurate fashion to real-time monitoring of the anatomical essential features of the hand.

MediaPipe Hands: A Baseline Framework of Feature Extraction:

An open-source application called MediaPipe Hands has been developed by Google and is the most crucial supporting technology of a current, real-time SLR. MediaPipe will not be unpopular as it is capable of tracking markersless hand movements on devices and is better-performing. It uses a highly effective pipeline, which involves two larger steps, palm detection model, which gives a bounding box of the hand, and a less heavy hand landmark model, which approximates the exact 3D location of 21 key anatomical joints (the skeletal structure) within the box.

The Need of the Coordinate Normalization:

The raw coordinates that are obtained using MediaPipe though accurate are the 3D landmark coordinates that depend not only on the distance of the user to the camera but also on the size of his physical hand. To obtain a strong and consistent downstream classifier, the data undergoes another significant preprocessing process known as coordinate normalization. It is normally accomplished by defining a fixed position point (like the centroid of the wrist or at the hand) as a reference point and specifying the location of all landmarks relative to some scale value of the size of the hand, like the palm width or length. This is an excellent measure to segregate the geometrical form of the gesture, and of the incidental spatial differences.

Input of Feature Engineering into the model:

The normalization process has the result of a fixed length vector that defines the pose of the hand a very compact and efficient representation of the sign. The geometric relationships of the various landmarks according to this calculation (e.g. the Euclidean distances or angles) may be further refined into a dense feature vector. This methodical feature engineering can only be possible because it transforms the visual data, which can be highly complex, into a simplified numerical format which deep learning models are most likely to consume.

Architectures of Static Sign Recognition CNN based:

The type of sign is highly vital in the type of classification structure that is embraced. The preferred architecture in the situation of a single sign (e.g. fingerspelling a specific letter) or isolated sign(s) is the convolutional Neural Networks (CNNs). CNNs perform well to acquire high-level hierarchical spatial patterns of normalized landmark vector of one frame. It has been demonstrated numerous times that the high-fidelity input of MediaPipe and the strong CNN classifiers work together to give the state of the art, and the accuracy level is 99.12 percent on standardized tests like ASL.

Distribution of CNNs and Networks:

The CNNs that are typically used in this task are the feature mapping convolutional layers, dimensionality reduction pooling layers and the fully connected layers where final classification is performed. ReLU activation functions are the standard using Dropout layers to enhance the learning results, and minimize the risks of the model overfitting, such that the network is able to achieve good performance on gestures it has not seen before.

Dynamic Sign Recognition using LSTM and RNNs:

By contrast, dynamic/continuous signals (e.g., word-level movement based gestures) require models with the ability to take into account the use of time. The reason behind this choice of Long Short-Term Memory (LSTM) networks, which is a particular type of Recurrent Neural Network (RNN), is that their internal gating mechanisms have the ability to store and manipulate information through sequences of input frames and therefore dynamically model the fluid trajectory of the hand movement over time.

CNN-LSTM Hybrid Architectures:

In order to take advantage of the spatial and temporal modeling potential, the hybrid CNN LSTM structures have been investigated in advanced research. In these designs, it trains a CNN to be a feature extractor on a frame per frame of a landmarks data and then the sequence of features is passed into an LSTM layer. This combination is especially effective in complex, word-level sign recognition as it is capable of capturing the exact shape of hands as well as the chronological development of the movement.

Concentrate on the Indian Sign Language (ISL) Subsets:

Although substantial effort has been made on the ASL, studies on the Indian Sign Language (ISL) and regional subsets are gaining more and more importance. Subsets of ISL have been targeted in projects, either using LSTM networks to deal with the sequentiality of dynamic signs. These works affirm the flexibility of the DL framework yet emphasize the enormous amount of vocabulary and regional differences that are yet to be given proper consideration.

Problems with Dataset Acquisition and Standardization:

One of the most crucial obstacles encountered by researchers, in particular, those who study regional sign languages, is the absence of standardized and publicly available large-scale datasets. The majority of successful SLR projects are based on self collected data, which creates an issue of benchmarking and cross-comparing the outcomes of various models and lines of research. The paramount requirement of model training is curated, regionally specific datasets of TSL.

The Emergent Multilingual and Regional Support:

The recurring theme in the existing literature is the blatant technological difference in favor of local and regional vernacular sign languages (RSLs). The research focus on major languages in the world (e.g., ASL) forms a digital divide and non-English speaking deaf communities are severely underserved. Such a shortfall requires a special emphasis on local initiatives to globalize access indeed.

In Other Indian RSL Recognition: Precedent:

The possibility of successful implementation of the MediaPipe paradigm and Deep Learning paradigm on the dialects of regions in India has been proven empirically. An interesting example is the research done on Assamese Sign Language, which was able to effectively use the MediaPipe framework to extract features and then use a custom-made classifier with a high accuracy rate. The paper is a good example of a methodological paradigm of dealing with other regional languages, such as TSL.

Focus on Live Time Performance and Efficiency:

To be useful in the real sense, an SLR system should be able to work in real time within a small latency range. This would require model optimization. With the current research focus is aimed at minimizing the complexity of the model (model slimming) and providing a high

inference rate, the system is viable to be used in casual communication that will not frustrate with the delays. One of the metrics is efficiency, which can be measured in combination with accuracy.

The End-to-End Communication Goal:
 Although recognition (gesture-to-text) is the major emphasis, full, two-way communication is the final objective that is expressed in numerous studies. This necessitates the addition of a Text-to-Speech (TTS) engine to the ultimate system so that the text being recognized can be transformed into audible speech output and hence a direct conversational loop between a hearing impaired user and a normal speaking user is achieved.

Portability and Mobile Deployment issues:
 In order to be as accessible as possible, the end system should be able to run on ubiquitous devices, and smartphones in particular. This moves the deployment platform out of the powerful desktop GPUs and into the limited mobile CPUs or embedded platforms. The studies should focus on the particular issues of streamlining models to run on mobile architecture (e.g. TensorFlow Lite) and creating a user-friendly interface to interact with on-the-go.

Justification of the Project: Within the Framework of Literature:
 The overall survey justifies the methodological decision of using MediaPipe to extract features and a CNN based model to classify the features, and it is the most successful and resource economical method to use according to the project needs. The fact that it was successfully applied in other RSLs is the additional evidence that the framework is adaptable to Indian sign variants.

Gaps in Existing Research:
 Although the technological maturity and high performance has been evident in the global SLR, there exists a deep technological gap in the field of regional and vernacular sign languages. Too much attention has been paid to the established global languages and communities such as the Tamil-speaking hearing-impaired population will not have special attention. An effective, available, and confirmed Tamil Sign Language (TSL) system, tested on a region-specific dataset and optimized to be used live has not been sufficiently discussed in the literature. This lack of localized, high-performance support is what makes the current project special in that it can fill this critical need in the region by implementing and customizing the proven MediaPipe and CNN framework to the linguistic and cultural needs of TSL.

Table					
Title	Author(s)	Year	Methodologies	Dataset(s)	Key findings
Real-Time Sign Language Recognition Using MediaPipe and Deep Learning Approaches: A Mobile Application Integration	Arjit Srivastava, Aditya Verma, Dr. Sadhana Rana, Er. Abhishek Dubey	2025	MediaPipe, Hybrid CNN-LSTM Networks	Not explicitly specified	Hybrid CNN-LSTM delivers superior performance for both static and dynamic signs. Focus on mobile application integration.
Enhancing Sign Language Detection through Mediapipe and Convolutional Neural Networks (CNN)	Gagandeep Singh, Aditya Raj Verma, Banavath RamJi, Karnim Meghwal, Praveen Kumar Dadheech	2024	MediaPipe, Convolutional Neural Networks (CNNs)	American Sign Language (ASL) datasets	Integration of CNN with MediaPipe provides high efficiency for real-time processing, achieving an accuracy of \$99.12\%\$ on ASL.
Real-time Assamese Sign Language Recognition using MediaPipe and Deep Learning	Jyotishman Bora, Saine Dehingia, Abhijit Boruah, Anuraag Anuj Chetia, Dikhit Gogoi	2023	MediaPipe, Deep Learning Classifier	Assamese Sign Language (RSL)	Successfully developed a technical approach for recognizing a vernacular sign language, addressing the need for localized tools.
RECOGNIZATION OF HAND GESTURES USING MEDIAPIPE HANDS	Kavana KM, Suma NR	2022	MediaPipe Hands (Palm Detector and Hand Landmark Model)	Not explicitly specified (General Hand Gestures)	Demonstrated a real-time, on-device hand tracking solution using MediaPipe to predict the hand skeleton, enabling touchless interaction.

SIGN LANGUAGE DETECTION USING MEDIAPIPE AND MACHINE LEARNING	Ms.Anjali Deshapande, Dr. Shashidhar Kini K	2024	MediaPipe (Feature Extraction), Machine Learning Algorithms	Varied dataset (General)	Aims to bridge communication gaps without the use of wearable sensors by leveraging MediaPipe for real-time feature extraction.
Real-Time Sign Language Recognition Using MediaPipe and Deep Learning Approaches: A Mobile Application Integration	Arjit Srivastava, Aditya Verma, Dr. Sadhana Rana, Er. Abhishek Dubey	2025	MediaPipe, Hybrid CNN-LSTM Networks	Not explicitly specified	Hybrid CNN-LSTM delivers superior performance for both static and dynamic signs. Focus on mobile application integration.

amiable solution that goes a long way in making digital access inclusive of the hearing and speech impaired community.

III. PROPOSED SYSTEM

The proposed system is an intelligent real time Tamil Sign Language (TSL) Recognition model which will be used to fill the gap in the communication of the hearing impaired community by converting the hand gestures into readable text. The architecture is designed with two advanced computer vision systems Mediapipe (hand tracking and feature extraction with very high precision) and Convolutional Neural Network (CNN) (gesture classification based on deep learning). Such integration is a seamless, end-to-end automated system that can execute a live system that anticipates gestures in real-time.

The process is triggered by the system with the live video input via a normal web camera. Through Mediapipe hand detector module, it recognizes 21 major hand points, which are referred to as the wrist, finger joints, and tips. After capture, the data is subject to a systematic preprocessing pipeline to make the model compatible and healthy (Module-2). These are Coordinate Normalization, in which the wrist is placed as the origin and palm length scaled landmarks are used to obtain size and position invariance. The fundamental features are then designed by calculating numerical values, that is, Euclidean distances, between the wrist and other landmarks to create a fixed-length vector.

To classify (Module-3), the system relies on a 1D Convolutional Neural Network (CNN) that is tailored to accommodate the study of spatial patterns and complex relationships among the feature vectors obtained by the hand landmarks. The network architecture consists of convolutional layers to identify features, pooling layers to minimize the data dimension and dropout layers to avoid overfitting of the model. The last thick layers carry out the classification and there is a Softmax output layer which gives the likelihood of each gesture class. The model is optimized with Adam and categorical cross-entropy is used as a loss (cat). Accuracy is defined as the main metric to measure the performance of the model. The data is randomly divided, 80 percent of which is used in training, and 20 percent in testing.

In its functionality, the integrated system takes the hand data of the user in real-time and runs it through the feature engineering pipeline before the final product, the resulting vector, is sent to the trained CNN model to predict the gesture instantly. When the classification is successful, the system shows the relevant Tamil letter or word on screen in Unicode font. This technology would be a very important tool of communication aid that is fast, precise and user friendly.

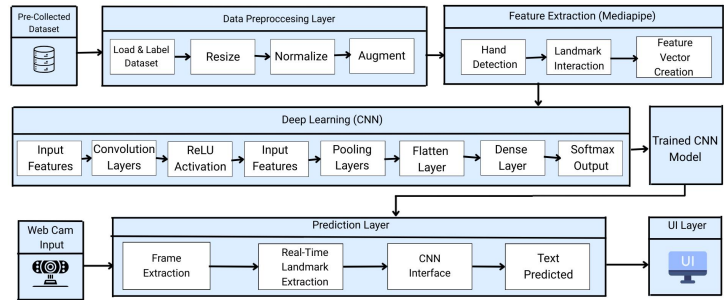


FIGURE I ARCHITECTURE DIAGRAM

IV CONCLUSION

The suggested AI-based realtime weapon and missing person detector system introduces the way in which the combination of deep learning and computer vision could significantly improve the security of the population and the efficiency of the surveillance. The system will combine a YOLOv10-based model with quick, precise weapon identification and Dlib based face recognition to identify missing individuals so as to offer a viable automated system of tracking dangers on a real-time basis. The lightweight nature of it makes it suitable in terms of performance with standard laptop-based hardware without using expensive graphics cards or complex infrastructure. The risky possibility or detection by remote awareness through e-mail notification in place of physical response using place-sound alarms is ensured to be communicated to the appropriate authorities in real-time.

The given framework can also address the current research gaps, including the absence of integration, real-time constraints, and the problem of false alarms, and offer the foundations upon which new studies in the area of intelligent surveillance systems and edge-based AI systems can be developed. The architecture will enable the possible expansion to IoT-enabled settings where notifications can be disseminated among linked devices or centralized monitoring tools. The suggested model, followed up by additional development of regional data expansion, explainable AI integration, and enhanced data privacy measures can be adjusted into a scalable,

transparent, socially responsible model of safety that is currently available to be implemented in smart cities, public infrastructure, and law enforcement activities.

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